

Matching for Causal Inference Without Balance Checking

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joint work with
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Preview Slide: Coarsened Exact Matching (CEM)

A very simple method of causal inference, with surprisingly powerful properties

- **Preprocess** (X, T) with CEM:

- ① Temporarily **coarsen** X as much as you're willing

- e.g., income ($\pm \$1,000$); education (high school, college, graduate); etc.
 - Most variables are pre-coarsened (e.g., men, women, wars, non-wars)
 - Easy to understand, or can be automated as for a histogram

- ② Perform **exact matching** on the coarsened X , $C(X)$

- Sort observations into strata, each with unique values of $C(X)$
 - Prune any stratum with 0 treated or 0 control units

- ③ **Pass on original (uncoarsened) units** except those pruned

- **(Optionally:** apply another matching method within strata)

- **Analyze** as without matching (adding weights for stratum-size)

⇒ A version of CEM: Last studied 40 years ago by Cochran

⇒ First used many decades before that

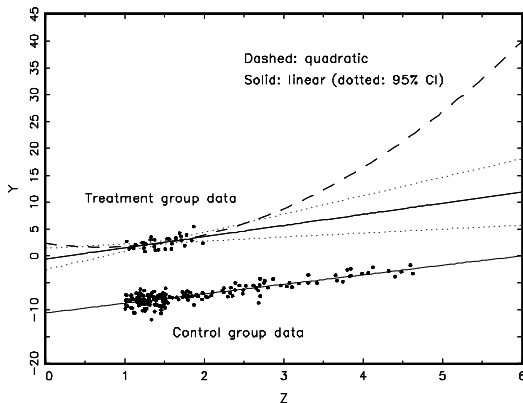
⇒ We prove: many new properties, uses, & extensions,
and show how it resolves many problems in the literature

Characteristics of Observational Data

- Plenty of data
- Data is of uncertain origin. Treatment assignment: not random, not controlled by investigator, not known
- Bias-Variance Tradeoff. **Bias:** huge concern; **Variance:** not so much
- **The idea of matching: sacrifice some data to avoid bias**
- Removing heterogeneous data will often **reduce variance** too
- (Medical experiments are the reverse: small- n with random treatment assignment; don't match unless something goes wrong)

Model Dependence

(King and Zeng, 2006: fig.4 *Political Analysis*)

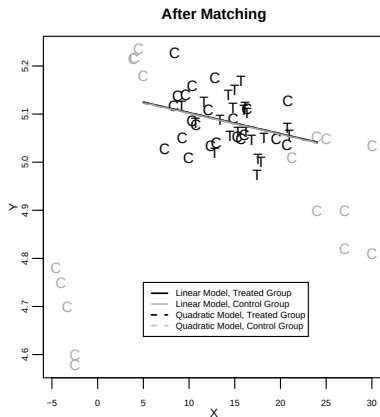
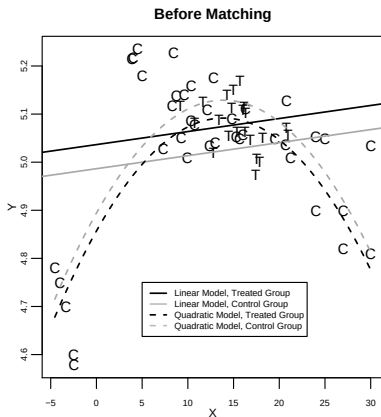


What to do?

- Preprocess I: Eliminate extrapolation region (a separate step)
- Preprocess II: Match (prune bad matches) within interpolation region
- Model remaining imbalance (same methods as without matching)

Matching within the Interpolation Region

(Ho, Imai, King, Stuart, 2007: fig.1, *Political Analysis*)



Matching reduces model dependence, bias, and variance

The Goals, with some more precision

- Notation:
 - Y_i Dependent variable
 - T_i Treatment variable (dichotomous)
 - X_i Covariates
- Treatment Effect for treated ($T_i = 1$) observation i :

$$\begin{aligned} \text{TE}_i &= Y_i(T_i = 1) - Y_i(T_i = 0) \\ &= \text{observed} - \text{unobserved} \end{aligned}$$

- Estimate $Y_i(T_i = 0)$ with Y_j from controls with X_j where $X_i \approx X_j$
- Sample Average Treatment effect on the Treated:

$$\text{SATT} = \frac{1}{n_T} \sum_{i \in \{T_i=1\}} \text{TE}_i$$

Matching Methods

- Goal: Find matching control units to estimate $Y_i(T_i = 0)$
- Exact matching ($X_i = X_j$): rarely works (except with lame X 's)
- Approximate matching methods ($X_i \approx X_j$):
 - Minimize scalar Mahalanobis distance between X_i and X_j
 - Propensity score: match on scalar $\pi = \text{logit}(T|X)$
 - Others: optimal, full, genetic, caliper, subclassification, etc.
- All methods attempt to achieve good **balance** between treated and control groups, so that X is unimportant

Problems With Existing Matching Methods

- Don't eliminate extrapolation region
- Don't work with multiply imputed data
- Most violate the congruence principle
- For most, the objective function is not optimized directly
- **Not well designed for observational data:**
 - Least important (variance): matched n **chosen ex ante**
 - Most important (bias): imbalance reduction **checked ex post**
- Hard to use: Improving balance on 1 variable can reduce it on others
 - Best practice: choose n -match-check, tweak-match-check, tweak-match-check, $\dots$
 - Actual practice: choose n , match, publish, STOP.
(Is balance even improved?)

Largest Class of Methods: Equal Percent Bias Reducing

- Goal: changing balance on 1 variable should not harm others
- For EPBR to be useful, it requires:
 - (a) X drawn randomly from a specified population $\mathbf{X}$,
 - (b) $\mathbf{X} \sim \text{Normal}$ (or DMPES)
 - (c) Matching algorithm is invariant to linear transformations of X .
 - (d) All X 's have the same importance w.r.t. Y
 - (e) Y is a linear function of X .
- EPBR Definition: Matched sample size set ex ante, and

$$\begin{array}{cc} \text{matched} & \text{original} \\ E(\bar{\mathbf{X}}_{m_T} - \bar{\mathbf{X}}_{m_C}) = & \gamma E(\bar{\mathbf{X}}_T - \bar{\mathbf{X}}_C) \end{array}$$

- When conditions hold:
 - Reducing mean-imbalance on one variable, reduces it on all
 - n set ex ante; balance calculated ex post
 - Methods can only be potentially EPBR (depends on the data)
 - EPBR controls only univariate expected mean-imbalance

A New Class of Methods: Monotonic Imbalance Bounding

- No restrictions on data types
- Designed for observational data (reversing EPBR):
 - Most important (bias): degree of balance **chosen ex ante**
 - Least important (variance): matched n **checked ex post**
- Balance is measured **in sample** (like blocked designs), not merely in expectation (like complete randomization)
- Covers **all forms of imbalance**: means, interactions, nonlinearities, moments, multivariate histogram, etc.
- **One** adjustable tuning parameter per variable
- **Convenient monotonicity property**: Reducing maximum imbalance on one X : no effect on others

MIB Formally:

$$D(X_{J,T}^{\epsilon}, X_{J,C}^{\epsilon}) \leq \gamma_J(\epsilon_J) \quad \text{vars to adjust (J)}$$

$$D(X_{H,T}^{\epsilon}, X_{H,C}^{\epsilon}) \leq \gamma_H(\epsilon_H) \quad \text{remaining vars (H)}$$

Treated and control X variables to adjust Remaining treated and control X variables “Imbalance” given chosen distance metric Bounds on

CEM as an MIB Method

- ϵ defines coarsening ($\epsilon = 0$ is exact matching)
- We Prove: CEM bounds the treated-control group difference, within strata and globally, for: means, variances, skewness, covariances, comoments, coskewness, co-kurtosis, quantiles, and full multivariate histogram.
- $\implies$ User sets ϵ and thus controls all multivariate differences, interactions, and nonlinearities, up to the chosen level
- Matched n is determined ex post
- If ϵ is set too large: you're left modeling remaining imbalances
- If ϵ is set too small: n may be too small
- What if ϵ is set appropriately, but n is still too small?
No magic method of matching can save you;
you're stuck modeling or collecting better data

Other CEM properties we prove

- Meets the **congruence principle**
 - The principle: data space = analysis space
 - Estimators that violate it are nonrobust and counterintuitive
 - CEM: ϵ_j is set using each variable's units
 - E.g., calipers (strata centered on each unit): would bin college drop out with 1st year grad student; and not bin Bill Gates & Warren Buffett
- **Automatically eliminates extrapolation region** (no separate step)
- **Approximate invariance to measurement error:**

	CEM	pscore	Mahalanobis	Genetic
% Common Units	96.5	70.2	80.9	80.0
- **Bounds model dependence** (not merely imbalance)
- **Bounds causal effect estimation error** (not merely model dependence)
- Computing: extremely **fast** and **memory-efficient** even for extremely large data sets; **can be fully automated**
- **Simple to teach**: coarsen, then exact match

CEM in Practice: EPBR-Compliant Data

Monte Carlo: $\mathbf{X}_T \sim N_5(\mathbf{0}, \Sigma)$ and $\mathbf{X}_C \sim N_5(\mathbf{1}, \Sigma)$. $n = 2,000$

Allow MAH & PSC to match with replacement; use automated CEM

Difference in means:

	X_1	X_2	X_3	X_4	X_5	Seconds
initial	1.00	1.00	1.00	1.00	1.00	
MAH	.20	.20	.20	.20	.20	.28
PSC	.11	.06	.03	.06	.03	.16
CEM	.04	.02	.06	.06	.04	.08

Local and multivariate imbalance:

	X_1	X_2	X_3	X_4	X_5	$\mathcal{L}_1$
initial						1.24
PSC	2.38	1.25	.74	1.25	.74	1.18
MAH	.56	.36	.29	.36	.29	1.13
CEM	.42	.26	.17	.22	.19	.78

⇒ CEM dominates EPBR-methods in EPBR Data

CEM in Practice: Non-EPBR Data

Monte Carlo: Exact replication of Diamond and Sekhon (2005), using data from Dehejia and Wahba (1999). CEM coarsening automated.

	BIAS	SD	RMSE	Seconds	$\mathcal{L}_1$
initial	-423.7	1566.5	1622.6	.00	1.28
MAH	784.8	737.9	1077.2	.03	1.08
PSC	260.5	1025.8	1058.4	.02	1.23
GEN	78.3	499.5	505.6	27.38	1.12
CEM	.8	111.4	111.4	.03	.76

⇒ CEM works well in non-EPBR data too

Extensions of CEM

- CEM and **Multiple Imputation for Missing Data**
 - ① put missing observation in stratum where plurality of imputations fall
 - ② pass on uncoarsened imputations to analysis stage
 - ③ Use the usual MI combining rules to analyze
- **Multicategory treatments**: No modification necessary; keep all strata with at least one value of each value of T
- **Blocking in Randomized Experiments**: no modification needed; randomly assign T within CEM strata
- **Automating user choices** Histogram bin size calculations, *Estimated* SATT error bound, Progressive Coarsening
- **Detecting Extreme Counterfactuals** (related to convex hull)
- **Improving Existing Matching Methods**:

Other methods applied within CEM strata inherit CEM properties

 - Propensity or Prognosis Score matching
 - Genetic Matching (now in GenMatch)
 - Synthetic Matching (Synth by Abadie, Diamond, & Hainmueller)
 - Nonparametric Adjustment (Abadie & Imbens)
 - & whatever else you all come up with.

For more information

<http://GKing.Harvard.edu/cem>